CS & IT ENGINEERING



Algorithms

Divide & Conquer



Lecture No.- 02

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Intro to DnC

Questions

Min-Max (Non-Dnc)

Topics to be Covered



- Topic
- Topic
- Topic

Min-Max

Binary Search

Pending Questions



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Min-Max Problem

Non-Divide & Conquer based

Total number of comparisons (Algo2)

Best → (n – 1)

Worst → 2*(n–1)

Avg → $\left(\frac{3n}{2} - 1\right)$

Algo2

```
if( )
{
}
else if( )
{
}
```

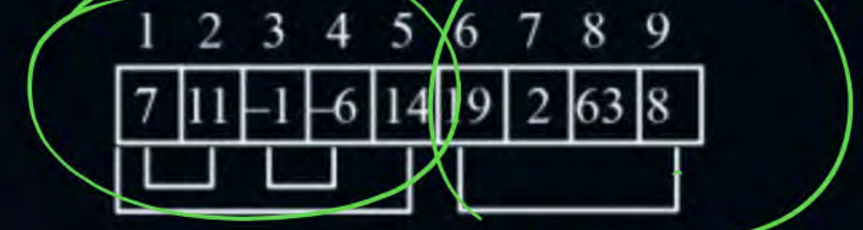
Algo1 →

```
if( )
{
}
if( )
{
}
```

B.C } 2*(n-1)
W.C }

Topic : Min-Max Problem

Divide & Conquer based



{ min = -6
  max = 63 }

m = ⌊ (start + end) / 2 ⌋

(1, 9, -6, 63)

min(-6, 2)

max(14, 63)

(start, end, min, max)

(1, 5, -6, 14) (6, 9, 2, 63)

(1, 3, -1, 11) (4, 5, -6, 14) (6, 7, 2, 19) (8, 9, 8, 63)

-6 14 | 19 2 | 63 8

(1, 2, 7, 11) (3, 3, -1, -1) -1

7 11

Small





Topic : Min-Max Problem

```
1.  Algorithm MaxMin(i, j, max, min)
2.  //a[1: n] is a global array. Parameters i and j are integers.
3.  {
4.  If (i=j) the max: min: = a[i]; // Small (P)
5.  else if (i=j–1) then // Another case of Small (P)
6.          {
7.  If (a[i] < a[j]) then
8.  {
9.  Max:=a[j]; min : = a[i];
10. }
```





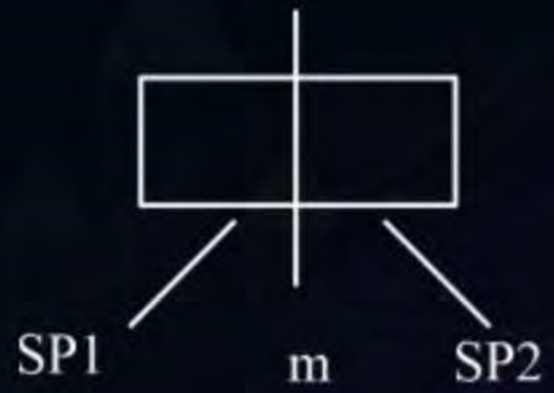


Topic : Min-Max Problem

```
13.         else
14.         {
15.             max: = a[i]; min : = a[j]
16.         }
17.   }
18.   else
19.   {     // If P is not small, divide P into subproblems.
20.         // Find where to split the set.
```

→ 17. } ⟶ small

18. else ⟶ Large (>2 elem)





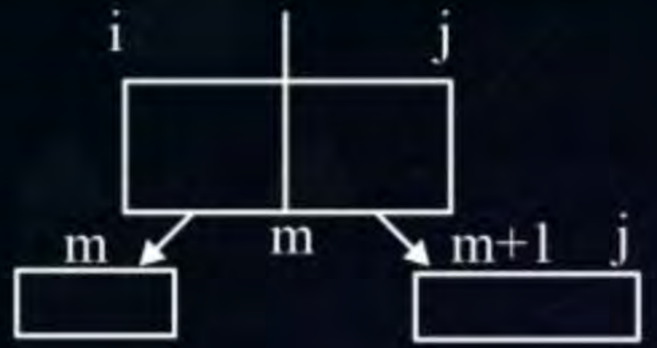
Topic : Min-Max Problem

```
21.  mid: = ⌊(i+j)/2⌋
22.  // Solve the subproblems.
23.  MaxMin (i, mid, max, min);
24.  MaxMin (mid + 1, j,max1, min1);
25.   // Combine the solution.
26.  If (max< max1) then max: =max1;
27.  If (min> min1) then min: = min1;
28.          }
29.  }
```

T(n/2)

T(n/2)

i = 1 + j / 2





Topic : Min-Max Problem

Performance of Divide & Conquer based

☆ Number of element comparisons:

Let T(n) → number of elem comparison reqd. for min-max on 'n' elem.

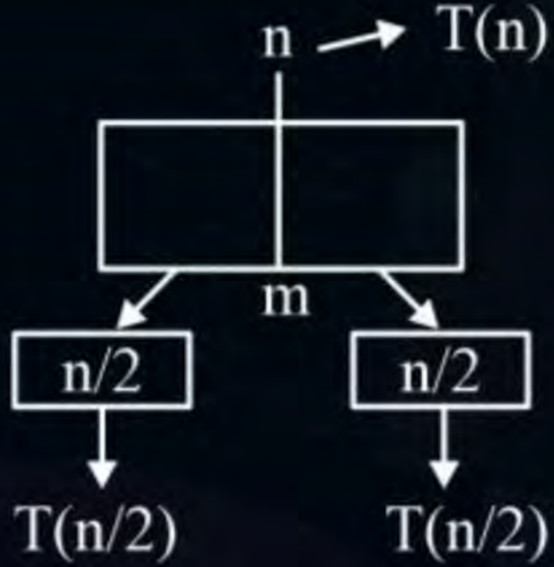
$T(n) = 0, n = 1$ → special case

$T(n) = 1, n = 2$ ✓

$T(n) = 2T(n/2) + 2, n > 2$



Value of Rec







Topic : Min-Max Problem

#Q. $T(n) = 2T(n/2) + 2$

$T(n/2) = 2T(n/2^2) + 2$

$T(n) = 2[2T(n/2^2) + 2] + 2$

$= 2^2T(n/2^2) + 2^2 + 2$

$= 2^2[2T(n/2^3) + 2] + 2^2 + 2$

$= 2^3T(n/2^3) + (2^3 + 2^2 + 2^1)$

General ~~then~~ term

$T(n) = 2^kT(n/2^k) + (2^k + 2^{k-1} \dots 2^1)$

$\Rightarrow T(n) = 2^kT(n/2^k) + \sum_{i=1}^{k} 2^i$

A (n/2 +1)

B (n–1)

C $\left(\frac{3n}{2} - 1\right)$

D $\left(\frac{3n}{2} - 2\right)$



Topic : Min-Max Problem

$2^1 + 2^2 - \ldots 2^k$

$$GP: \Rightarrow \left.\begin{matrix} a = 2 \\ r = 2 \\ n = k \end{matrix}\right\} \frac{a(r^n - 1)}{r-1}$$

$$= \frac{2(2^k - 1)}{2-1} = \boxed{2^k - 2} \quad = a(r^k - 1) = 2 * 2^k - 2$$



Topic : Min-Max Problem

$$T(n) = 2^k T(n/2^k) + 2(2^k - 1)$$

For BC

$n/2^k = 2$

$n = 2 \times 2^k$

$n = 2^{(k+1)}$

$(k+1) = \log_2 n$

$$\Rightarrow \frac{n}{2} \times T(2) + 2\left(\frac{n}{2} - 1\right)$$

$$\Rightarrow \frac{n}{2} \times 1 + n - 2$$

$$\Rightarrow \frac{n}{2} + n - 2 = \boxed{\frac{3n}{2} - 2}$$ ☆☆

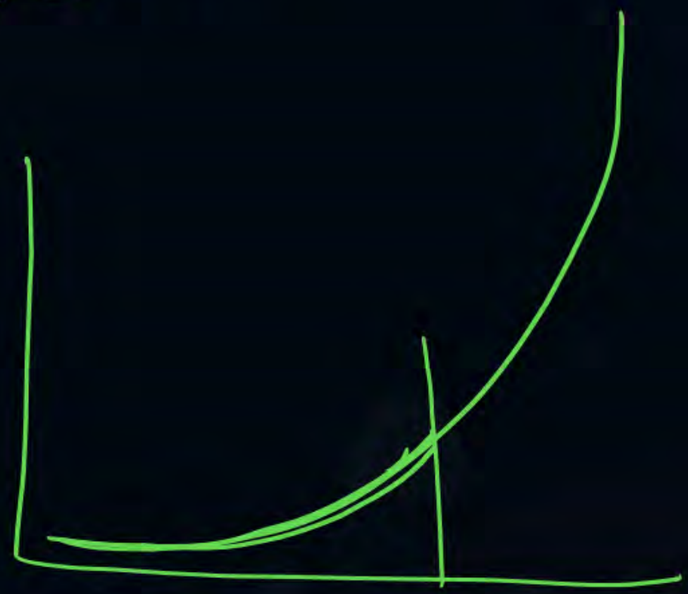
$\left(\frac{3n}{2} - 1\right)$



Topic : Min-Max Problem

Summary: DnC vs D&C → Min-Max Algo:

	Non-DnC (Algo2)	DnC	
Best case Algo2 Decr order	(n–1)	$\left(\frac{3n}{2}-2\right)$	
Wrost case Algo2 Incr order	2*(n–1)	$\left(\frac{3n}{2}-2\right)$	
Avg case (Random order)	$\left(\frac{3n}{2}-1\right)$	$\left(\frac{3n}{2}-2\right)$	





Topic : Min-Max Problem

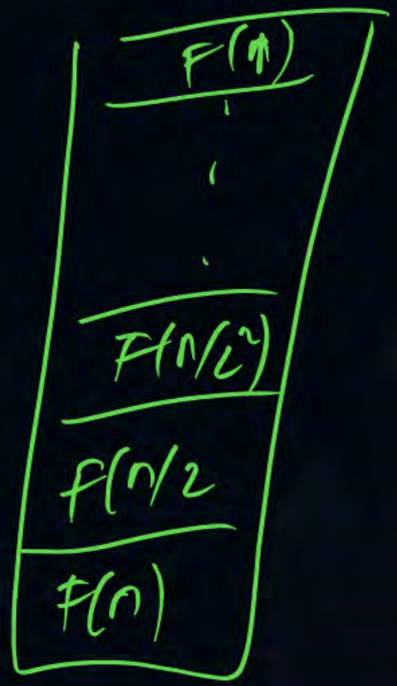
Space complexity:-

Non-D&C (Algo2) $\rightarrow 0(1)$ ✓

D&C $\rightarrow$ Recrsion stak $\rightarrow 0(\log_2 n)$

max = INT_min

min = INT_max





Topic : Binary Search

Binary Search:

1

Searching
- 1) Linear
- 2) Binary



Topic : Binary Search

#Q. Given a Binary Tree of 'n' no des,
min. Height (depth)
max. Height (depth)

Level → 1 → Root

0 — 2

50



1 — 10

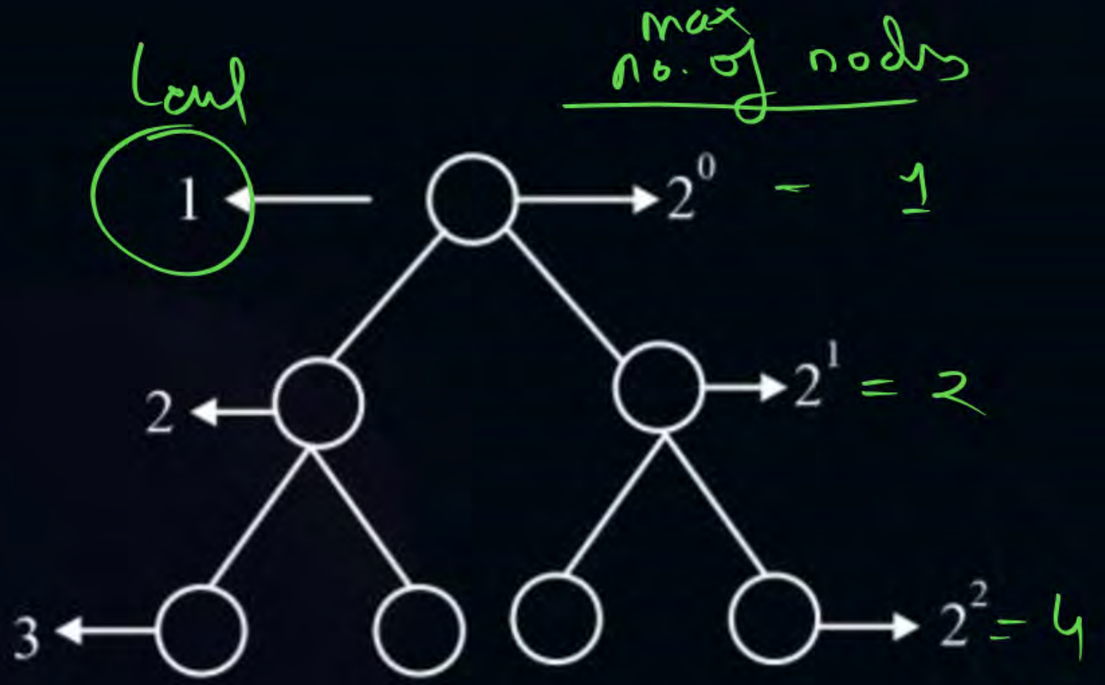
Boxes | 10 | → min = 5

Box | 1 |



Topic : Binary Search

Level:-



$2^{(i-1)}$



Topic : Binary Search

Max number of nodes at any level 'i' in BT = $2^{(i-1)}$

(Root → Li)





Topic : Binary Search

#Q. Total number nodes in a full Binary Tree of height h.

$$= 2^{1-1} + 2^{2-1} + 2^{3-1} \ldots + 2^{h-1}$$

$$= \sum_{i=1}^{h} 2^{(i-1)} = \sum_{i-1}^{h} \frac{2^i}{2} = \frac{1}{2} * \sum_{i=1}^{h} 2^i$$

Total nodes:

$$n = \frac{2(2^h-1)}{2-1} \times \frac{1}{2}$$

$$n = 2(2^h - 1) \times \frac{1}{2}$$

$$n = 2^h - 1$$

$$2^h = n + 1$$

$$h = \log_2(n+1) \approx h = 0(\log_2 n)$$



Topic : Binary Search

Max-Height

Level

Level:

emery level 1 elem

Height h

Total nodes = $\sum_{i=1}^{h} 1$

$$n = \frac{(1 + 1 + \ldots 1)}{\text{h times}}$$

$n = h \Rightarrow h = n$

Max-height of a BT

$H = O(n)$

max





Topic : Binary Search

Range of height of a Binary Tree.

Level start at 1

$$\log_2^n \leq h \leq n$$



Topic : Min-Max Problem

famous

#Q. Given a Binary Tree
level starts at 1
every level i has exactly 'i' nodes
Height of such a BT = ?

A n^2

B $\sqrt{n}$ → 34%

C n → 28%

D logn → 28%



Topic : Design Strategies

Height = h

Eg. Level i → i nodes

Total nodes = $\sum_{i=1}^{h} i$ $\quad 1+2+3 \cdots h$

$$n = \frac{h(h+1)}{2}$$

$$n = \frac{h^2+h}{2}$$

$n = O(h^2)$

$h = \sqrt{n}$ $\quad h^2 = O(n)$

Level





Topic : Divide and Conquer

#Q. for i →(1 to (n – 1))

for j →(i + 1 to n)

for k → (1 to j)

print (“*”)

Imp Logic

$$\underset{i=1}{\sum_{k=2}^{n} k} + \underset{i=2}{\sum_{k=3}^{n} k} + \underset{i=3}{\sum_{k=4}^{n} k} \cdots\cdots \sum_{k=n}^{n} k$$

$$\left[\sum_{k=1}^{n} k - 1\right] + \left[\sum_{k=1}^{n} k - (1+2)\right] + \left[\sum_{k=1}^{n} k - (1+2+3)\right] \cdots\cdots \text{(n-1) times.}$$

$$= \left(\sum_{k=1}^{n} k\right) * (n-1) - \left[(1) + (1+2) + (1+2+3) \cdots (1+2+3 \cdots (n-1))\right]$$

$$= \frac{n(n+1)}{2} * (n-1) - \left[\sum_{j=1}^{(n-1)} \frac{j(j+1)}{2}\right]$$

Doubt

$$(1) + (1+2) + (1+2+3) + \cdots \left[1+2+3 \cdots (n-1)\right]$$

$$\sum_{1}^{1} j \qquad \sum_{1}^{2} j \qquad \sum_{1}^{3} j \qquad \sum_{1}^{n-1} j$$

$$\sum_{j=1}^{n-1} \frac{j(j+1)}{2}$$

Sorting : Imp Questions



Topic : Bubble Sort

(P.Y.Q)

Apti + Algo

#Q. If all permutations are equally likely, what is the expected number of inversions in a randomly chosen permutation of 1.....n?.

40%

A $\frac{n(n-1)}{2}$

B $\frac{n(n-1)}{4}$

C $\frac{n(n+1)}{4}$

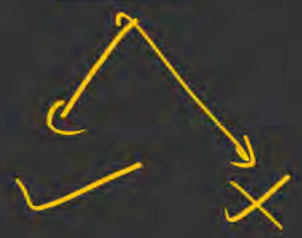
D $2n[\log_2 n]$

Soln: n elems

$\hookrightarrow {}^{n}C_2$ pairs

$$= \frac{n!}{(n-2)! \times 2!} = \frac{n(n-1)\times(n-2)!}{(n-2)! \times 2} = \left[\frac{n(n-1)}{2}\right] \text{pairs}$$

(a, b)



$$P(\text{Pair} = \text{inversion}) = \frac{1}{2}$$

$$\text{Expected no. of inversions} = \#\text{pairs} * P(\text{pair} = \text{inv})$$

$$= \frac{n(n-1)}{2} * \frac{1}{2} = \boxed{\frac{n(n-1)}{4}}$$



Topic : Bubble Sort

V. Imp

#Q. What would be the worst case time complexity of the insertion sort algorithm, if the inputs are restricted to permutation of 1... n with at most n inversions ?

A $\Theta(n^2)$

B $\Theta(n \log n)$

C $\Theta(n^{1.5})$

D $\Theta(n)$

40%

✗ overall W.C

| 10 | 9 | 7 | 5 |

inv

$\Rightarrow \frac{n(n-1)}{2}$

✓ given W.C

| 2 | 5 | 9 | 1 |

$\Rightarrow$ inv $\rightarrow (n-1)$

at max $\longrightarrow$ $O(n)$ inv

TC: $O(n+d)$

$d = O(n)$

$TC = O(n+n) = \underline{\underline{O(n)}}$



Topic : Bubble Sort

H.W

$(n^{k/2}, n^k)$

#Q. If we use Radix Sort to sort n integers in the range ~~$[n^{k/2}, n^k]$~~, for some $k > 0$ which is independent of n, the time taken would be:

A $\Theta(n)$

B $\Theta(kn)$

C $\Theta(n \log n)$

D $\Theta(n^2)$



Topic : Bubble Sort

#Q. Give the correct matching for the following pairs :

List-I	List-II
A. O(log n)	P. Selection
B. O(n)	Q. Insertion sort
C. O(n log n)	R. Binary search
D. $O(n^2)$	S. Merge sort

Codes:

A – R B – P C – Q D – S

B – R B – P C – S D – Q

C – P B – R C – S D – Q

D – P B – S C – R D – Q



Topic : Bubble Sort

H.W

#Q. Assume that the algorithm considered here sort the input sequence in ascending order. If the input is already in ascending order, which of the following are TRUE ?

I. Quick sort runs in $\Theta(n^2)$ time.
II. Bubble sort runs in $\Theta(n^2)$ time.
III. Merge sort runs in $\Theta(n)$ time.
IV. Insertion sort runs in $\Theta(n)$ time.

A. I and II only

B. I and III only

C. II and IV only

D. I and IV only



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Min - Max

Intro to Binary Search.



THANK - YOU